

9 Splines

We have the non-parametric regression situation

$$y = f(x) + \epsilon$$

with data points $(x_1, y_1, \dots, (x_n, y_n)$. The idea of spline is we want to find a function $f(x)$ that fits the data well but also is very smooth.

9.1 Cubic splines

Define a set of knots $\xi_1, \dots, \xi_k$. We want the function f to

- Be a cubic polynomial between every pair of knots
- Be continuous at each knot.
- Have continuous first and second derivatives at each knot.

It turns out, we can write f in terms of $k + 3$ basis functions:

$$f(x) = \beta_0 + \beta_1 x = \beta_2 x^2 + \beta_3 x^3 + \beta_4 h(x, \xi_1) + \dots + \beta_{k+3} h(x, \xi_k)$$

where $h(a, b) = (a - b)^3 I(a > b)$. To find f , solve for betas that minimize the sum of squares.

Spline which is linear instead of cubic for $x < \xi_1$ and $x > \xi_k$ are called natural cubic splines. The locations of the knots are typically quantiles of X . The number of knots, k , is chosen by cross validation.

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> lines(smooth.spline(x=X,y=Y,spar=0.2))  
> lines(smooth.spline(x=X,y=Y,spar=0.8))
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9.2 Smoothing Splines

We want to find a function $f(x)$ such that

$$\sum_{i=1}^n (Y_i - f(x_i))^2 + \lambda \int_{x_{(1)}}^{x_{(n)}} |f''(x)|^2 dx$$

is minimized. The quantity λ is called smoothness penalty/parameter that determines how much we will lose if we allow the function to adapt to the data. Without this part of penalty, the best function f will be the ones passing through every observation. The minimizer is a natural cubic spline. Choose λ by cross validation.